UNIT III · CHEAT SHEET

Drone technology, propulsion and business use

Terminology, generations, power sources, rotor physics, payloads, positioning, use cases and ROI

Words that get marks

- **UAS / RPAS / sUAS:** whole system / licensed remote pilot (ICAO) / ≤ 25 kg baseline.
- **AGL, MSL, HAE:** above ground, above mean sea level, above the ellipsoid (GNSS).
- **Geofence, failsafe, RTL/RTH, lost link:** see Unit V sheet.

How drones got here

- Analog RC + nitro engines → MEMS gyros and accelerometers → digital telemetry → brushless motors → LiPo batteries → GNSS after Selective Availability ended (May 2000) → RTK → open-source autopilots → smartphone-scale electronics → edge AI, 5G, UTM.

Power sources (specific energy)

- **LiPo / Li-ion:** 200–260 Wh/kg; 75–85% efficient; 20–60 min.
- **H_2 fuel cell (system):** 500–800 Wh/kg; 2–4 h; zero vibration.
- **Series hybrid (system):** 800–1,500 Wh/kg; 2–5 h; heavy-lift multirotors.
- **Gasoline / heavy fuel:** 10,000–12,000 Wh/kg but 20–34% efficient; vibration, noise.
- Series hybrid: engine → generator → battery buffer → motors. Parallel: engine and motor both drive the prop.

Rotor physics

- **Inflow:** $v_i = \sqrt{(T/(2\rho A))}$; **disk loading:** $DL = T/A = 4T/(\pi D^2)$
- **Ideal hover power:** $P_i = T^{1.5}/\sqrt{(2\rho A)} = T \cdot v_i$
- **Scaling:** $T = C_T \cdot \rho \cdot n^2 \cdot D^4$; $P = C_P \cdot \rho \cdot n^3 \cdot D^5$
- **Reaction torque:** $Q = (C_P/2\pi) \cdot \rho \cdot n^2 \cdot D^5$; CW/CCW pairs cancel it; yaw comes from unbalancing it.
- Example: 10 N rotor, D = 254 mm → 89.7 W; D = 127 mm → 179.5 W.

Payloads

- **RGB:** maps, photogrammetry. **Multispectral:** crop health.
- **Thermal (radiometric):** hot spots, solar, leaks. **LiDAR:** terrain under vegetation.
- **GSD = p·H/F.** p = 2.4 µm, H = 100 m, F = 8.8 mm → 2.7 cm/px.

RTK vs PPK

- **RTK:** base sends corrections live (radio/NTRIP); 1–2 cm while the link holds.
- **PPK:** both log raw GNSS (RINEX); processed after landing; immune to dropouts.
- EKF fuses IMU + baro + GNSS; optical flow / VIO when GNSS is weak.

Use cases

- Precision agriculture: NDVI = (NIR − Red)/(NIR + Red); stress map → variable-rate spraying.
- Inspection: civil structures, power lines, solar and wind.
- Survey and mapping (SfM, 75–85% front, 65–75% side overlap); construction/BIM; mining stockpile volumes; logistics and medical delivery.

Business value

- Four drivers: cost, time, safety, data quality.
- **ROI** = (Benefit − Cost)/Cost × 100%. **Payback** = capital ÷ annual net savings.
- Illustrative: ₹12 lakh capital, ₹22 lakh/yr savings → payback 0.55 yr; first-year ROI 83%.
- Always state what "cost" includes (capital, operating, or both).

Exam traps

- Comparing fuel and battery on Wh/kg without efficiency.
- Using airspeed in place of ground speed; using D^2 for thrust (it is D^4 at fixed n).

Source: course reader (Chapters 1–5)